

(Q) Implement the Boolean function $F(A,B,C)=\Sigma(1,3,5,6)$ with using 4*1 multiplexer.

- Drawing truth table according to the question.

INPUTS				OUTPUTS
MINTERM	A	B	C	Y
0	0	0	0	0
1	0	0	1	1
2	0	1	0	0
3	0	1	1	1
4	1	0	0	0
5	1	0	1	1
6	1	1	0	1
7	1	1	1	0

- Implementation table for 4*1 MUX.

A	B	C	Y	Data IN
0	0	0	0	C
0	0	1	1	
0	1	0	0	C
0	1	1	1	
1	0	0	0	C
1	0	1	1	
1	1	0	1	C'
1	1	1	0	

8 line to 3 line Encoder



Truth Table

D0	D1	D2	D3	D4	D5	D6	D7	A	B	C
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	1
0	0	0	0	0	0	1	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1

8 line to 3 line Encoder

Truth Table

D0	D1	D2	D3	D4	D5	D6	D7	A	B	C
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	1
0	0	0	0	0	0	1	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1

$$A = D4 + D5 + D6 + D7$$

$$B = D2 + D3 + D6 + D7$$

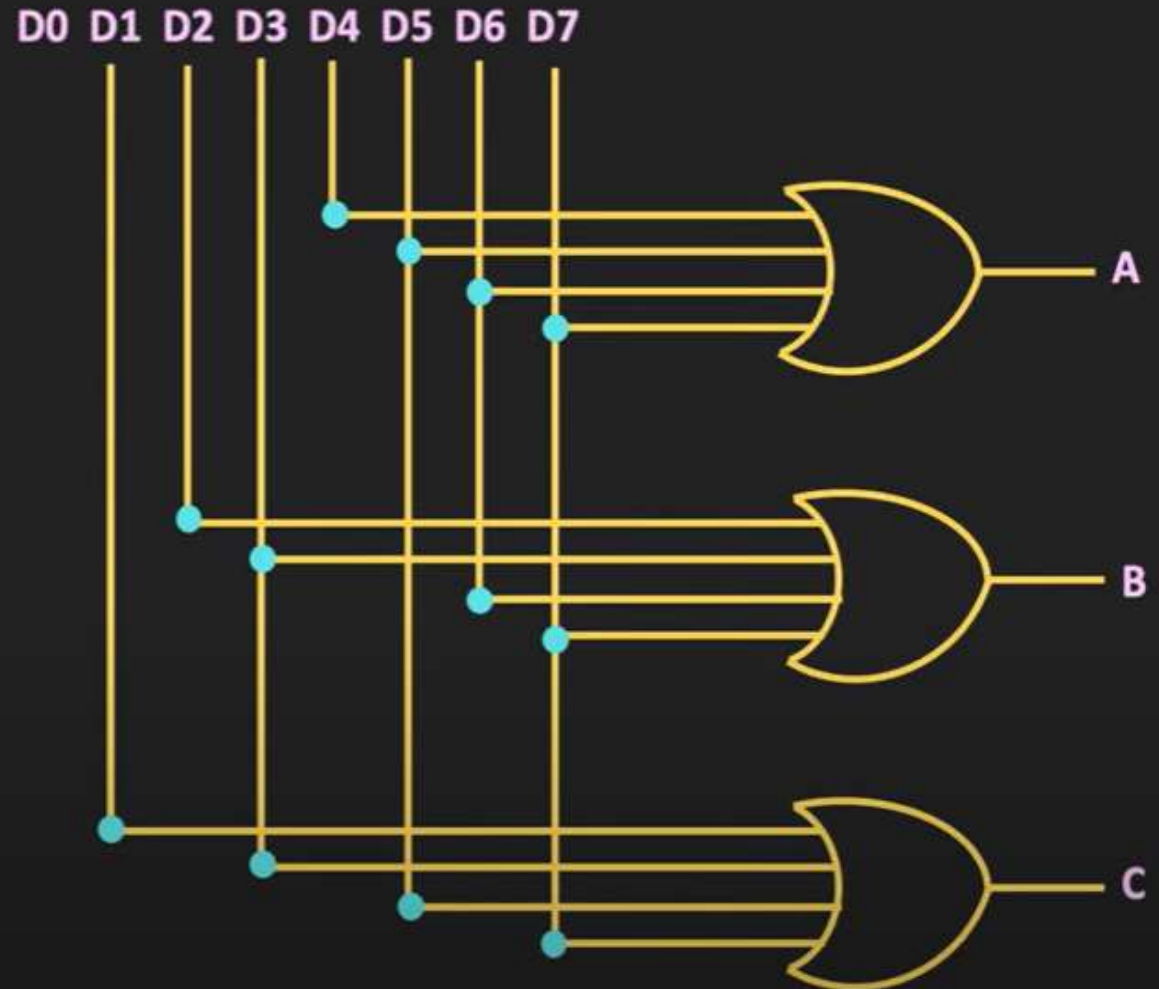
$$C = D1 + D3 + D5 + D7$$

8 line to 3 line Encoder

$$A = D4 + D5 + D6 + D7$$

$$B = D2 + D3 + D6 + D7$$

$$C = D1 + D3 + D5 + D7$$



Decimal to BCD Encoder



Decimal to BCD Encoder

Truth Table

D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	A	B	C	D
1	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	0	0	0	1	0	0
0	0	0	0	0	1	0	0	0	0	0	1	0	1
0	0	0	0	0	0	1	0	0	0	0	1	1	0
0	0	0	0	0	0	0	1	0	0	0	1	1	1
0	0	0	0	0	0	0	0	1	0	1	0	0	0
0	0	0	0	0	0	0	0	0	1	1	0	0	1

$$A = D8 + D9$$

$$B = D4 + D5 + D6 + D7$$

$$C = D2 + D3 + D6 + D7$$

$$D = D1 + D3 + D5 + D7 + D9$$

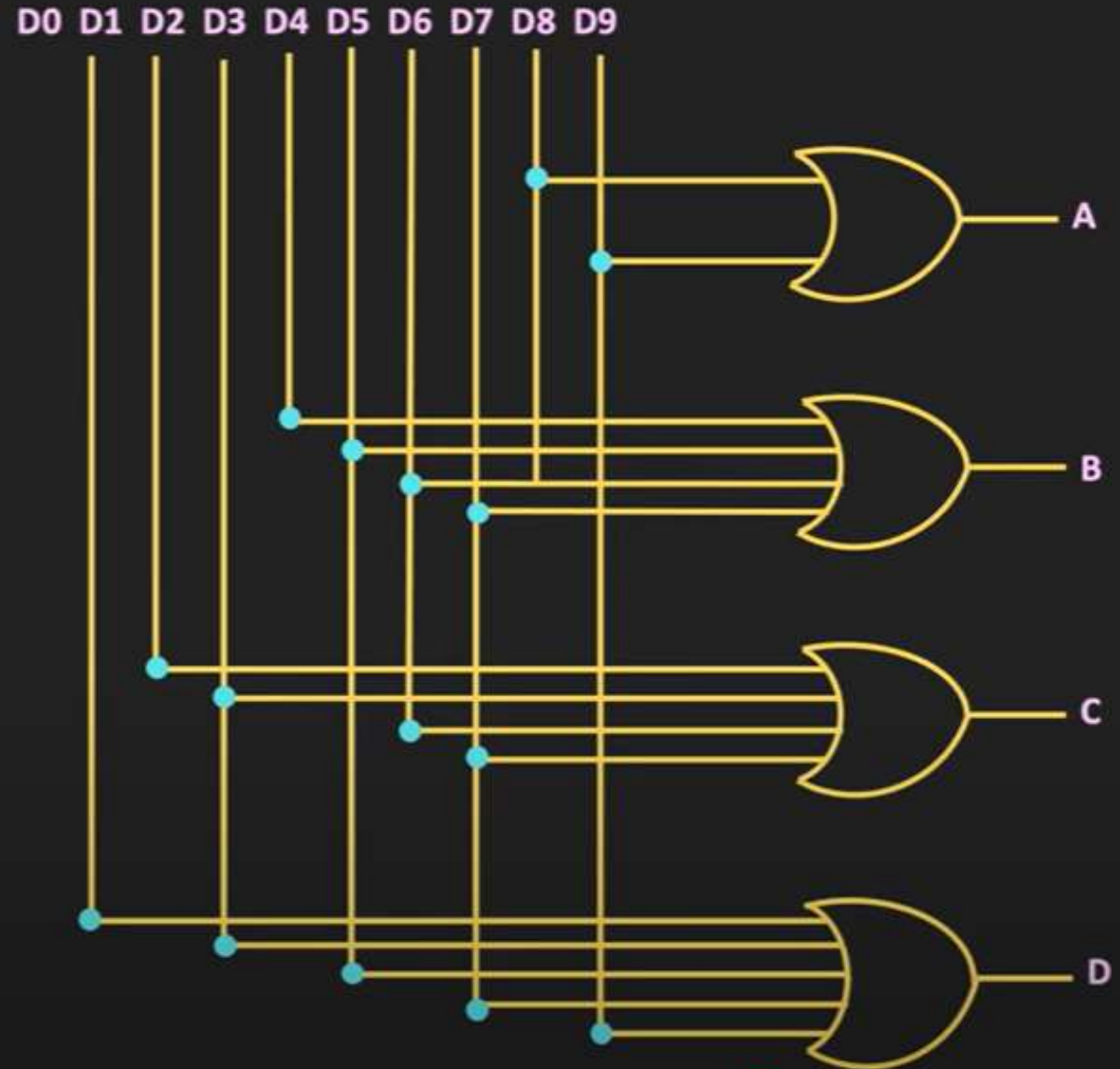
Decimal to BCD Encoder

$$A = D_8 + D_9$$

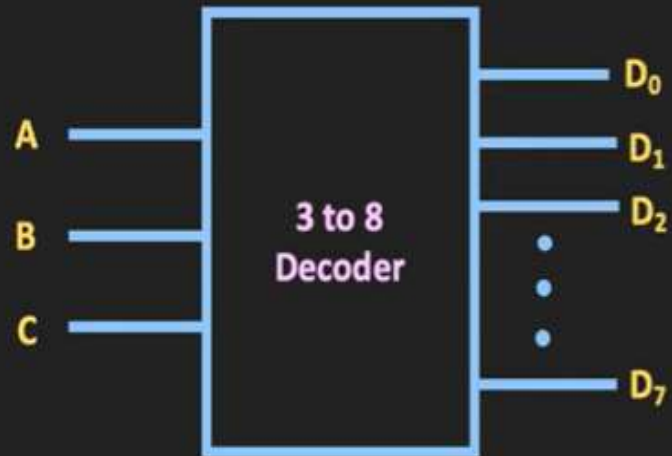
$$B = D_4 + D_5 + D_6 + D_7$$

$$C = D_2 + D_3 + D_6 + D_7$$

$$D = D_1 + D_3 + D_5 + D_7 + D_9$$



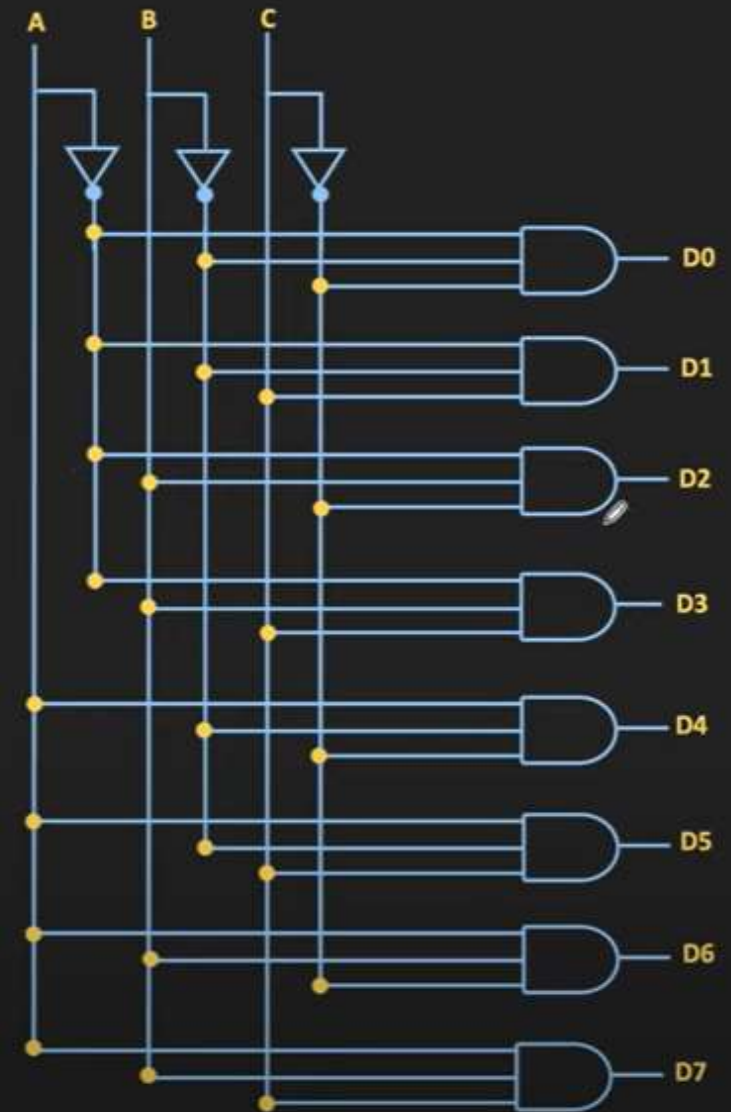
3 to 8 Decoder



Truth Table

A	B	C	D ₀	D ₁	D ₂	D ₃	D ₄	D ₅	D ₆	D ₇
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1

3 to 8 Decoder



BCD to Decimal Decoder

Truth Table

[illegible]

8 line to 3 line Encoder



Truth Table

D0	D1	D2	D3	D4	D5	D6	D7	A	B	C
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	1
0	0	0	0	0	0	1	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1

8 line to 3 line Encoder

Truth Table

D0	D1	D2	D3	D4	D5	D6	D7	A	B	C
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	1
0	0	0	0	0	0	1	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1

$$A = D4 + D5 + D6 + D7$$

$$B = D2 + D3 + D6 + D7$$

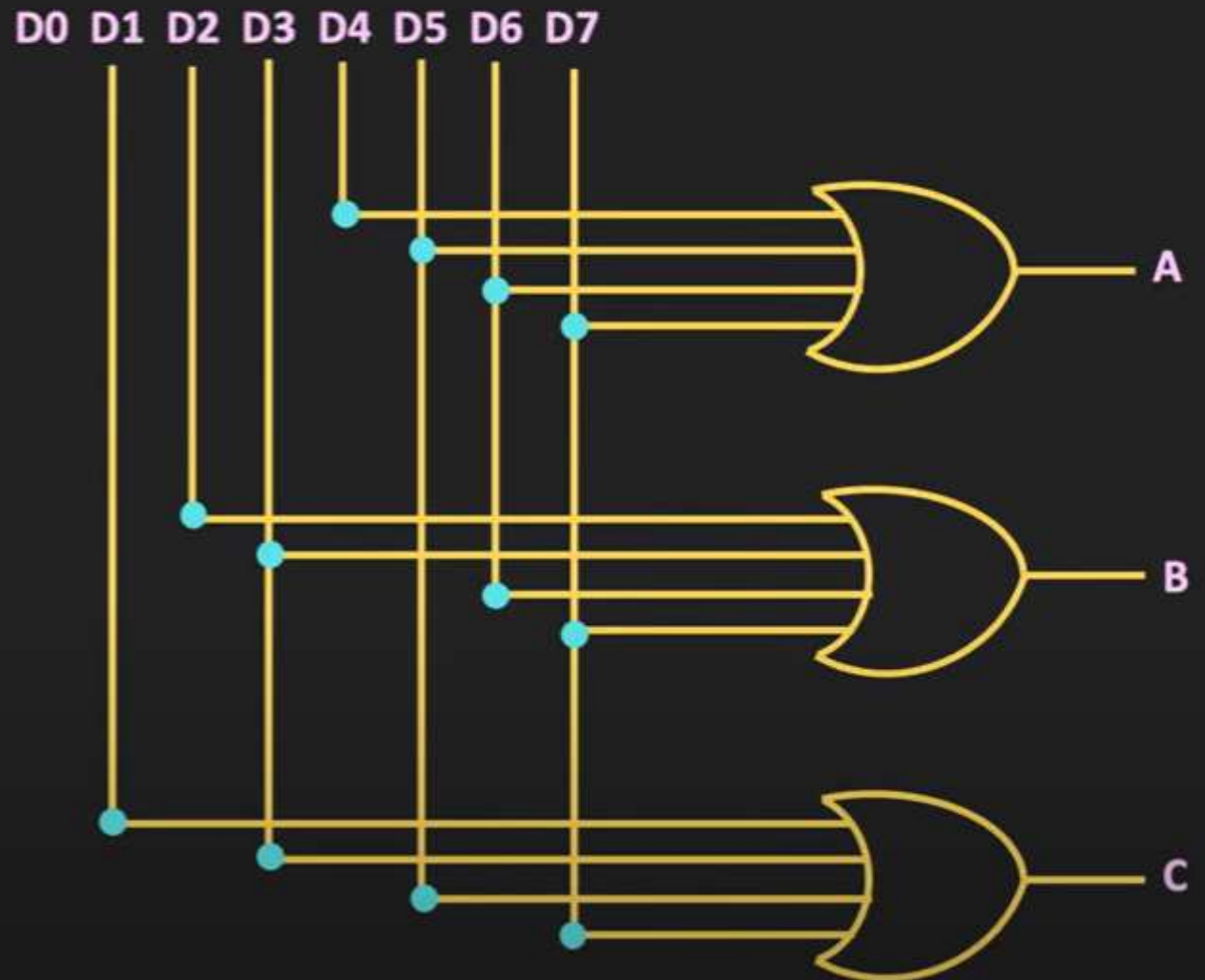
$$C = D1 + D3 + D5 + D7$$

8 line to 3 line Encoder

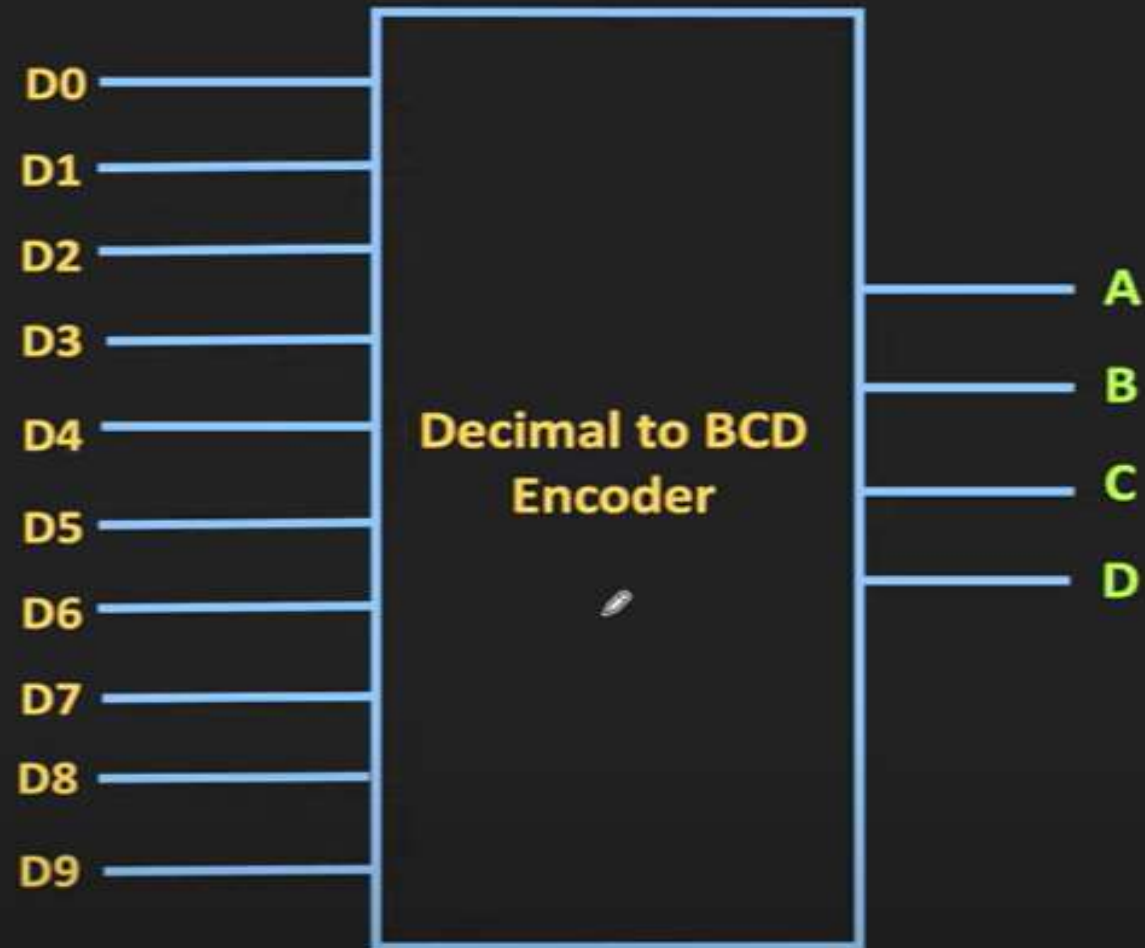
$$A = D4 + D5 + D6 + D7$$

$$B = D2 + D3 + D6 + D7$$

$$C = D1 + D3 + D5 + D7$$



Decimal to BCD Encoder



Decimal to BCD Encoder

Truth Table

D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	A	B	C	D
1	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	0	0	0	1	0	0
0	0	0	0	0	1	0	0	0	0	0	1	0	1
0	0	0	0	0	0	1	0	0	0	0	1	1	0
0	0	0	0	0	0	0	1	0	0	0	1	1	1
0	0	0	0	0	0	0	0	1	0	1	0	0	0
0	0	0	0	0	0	0	0	0	1	1	0	0	1

$$A = D8 + D9$$

$$B = D4 + D5 + D6 + D7$$

$$C = D2 + D3 + D6 + D7$$

$$D = D1 + D3 + D5 + D7 + D9$$

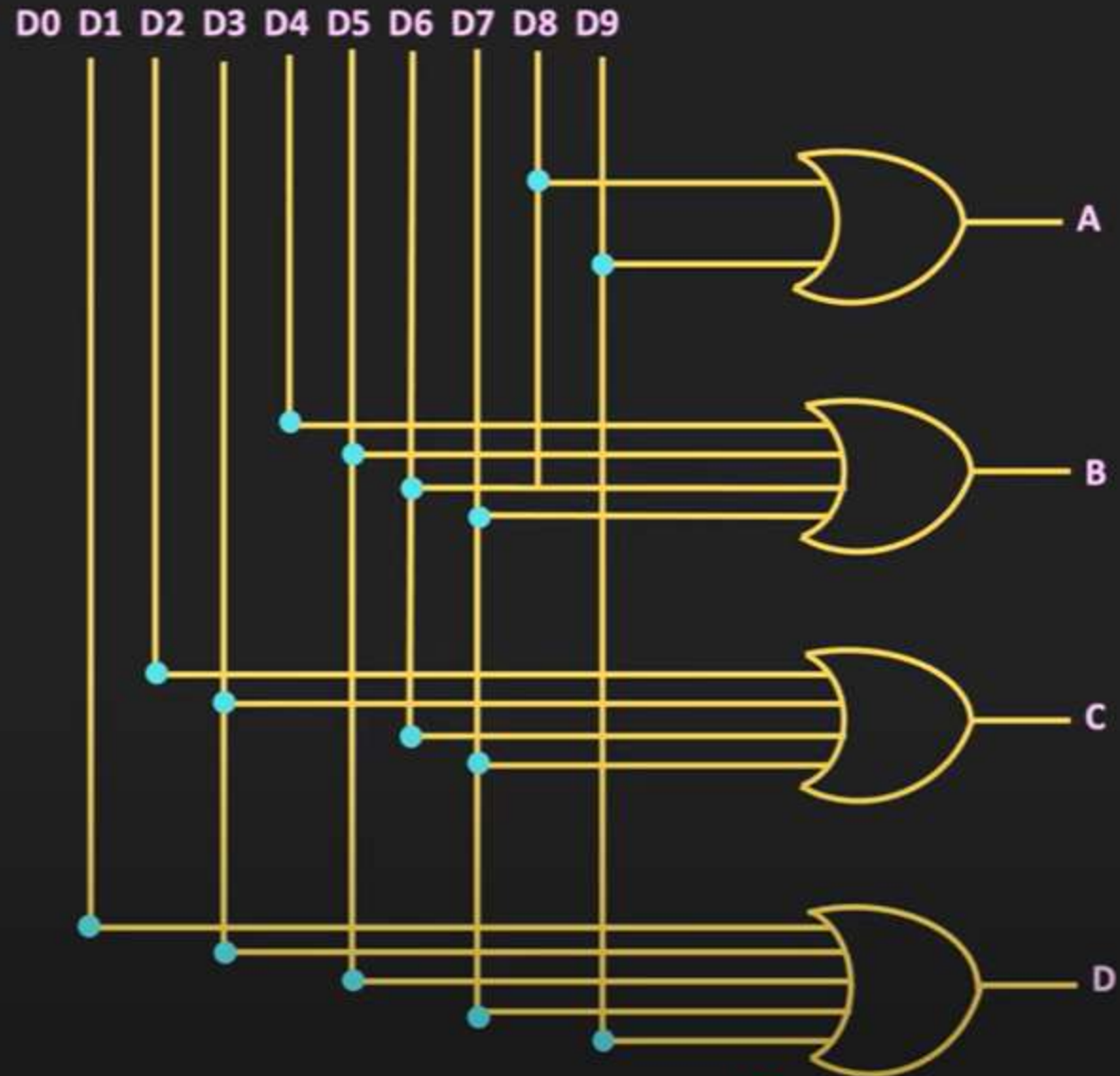
Decimal to BCD Encoder

$$A = D_8 + D_9$$

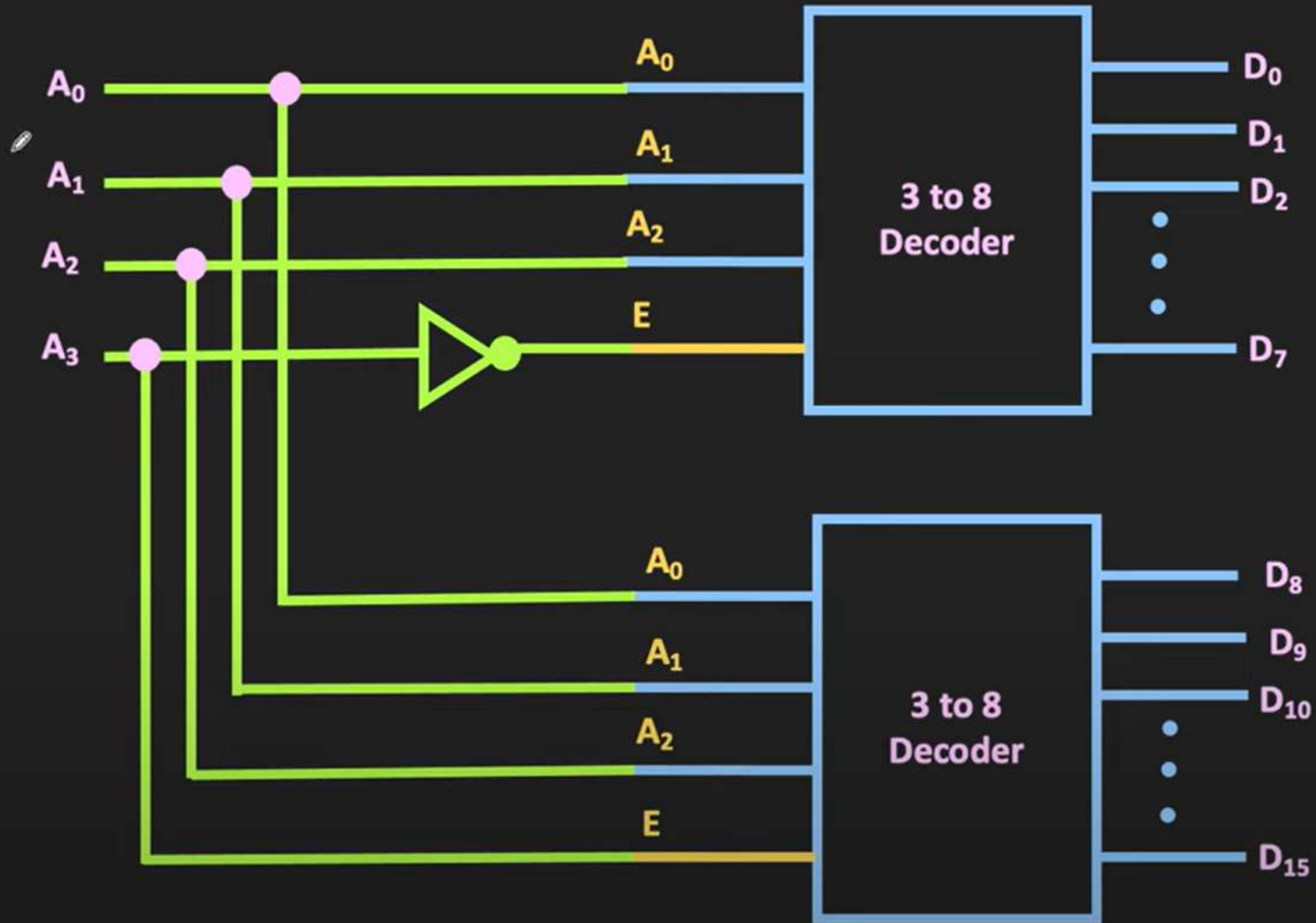
$$B = D_4 + D_5 + D_6 + D_7$$

$$C = D_2 + D_3 + D_6 + D_7$$

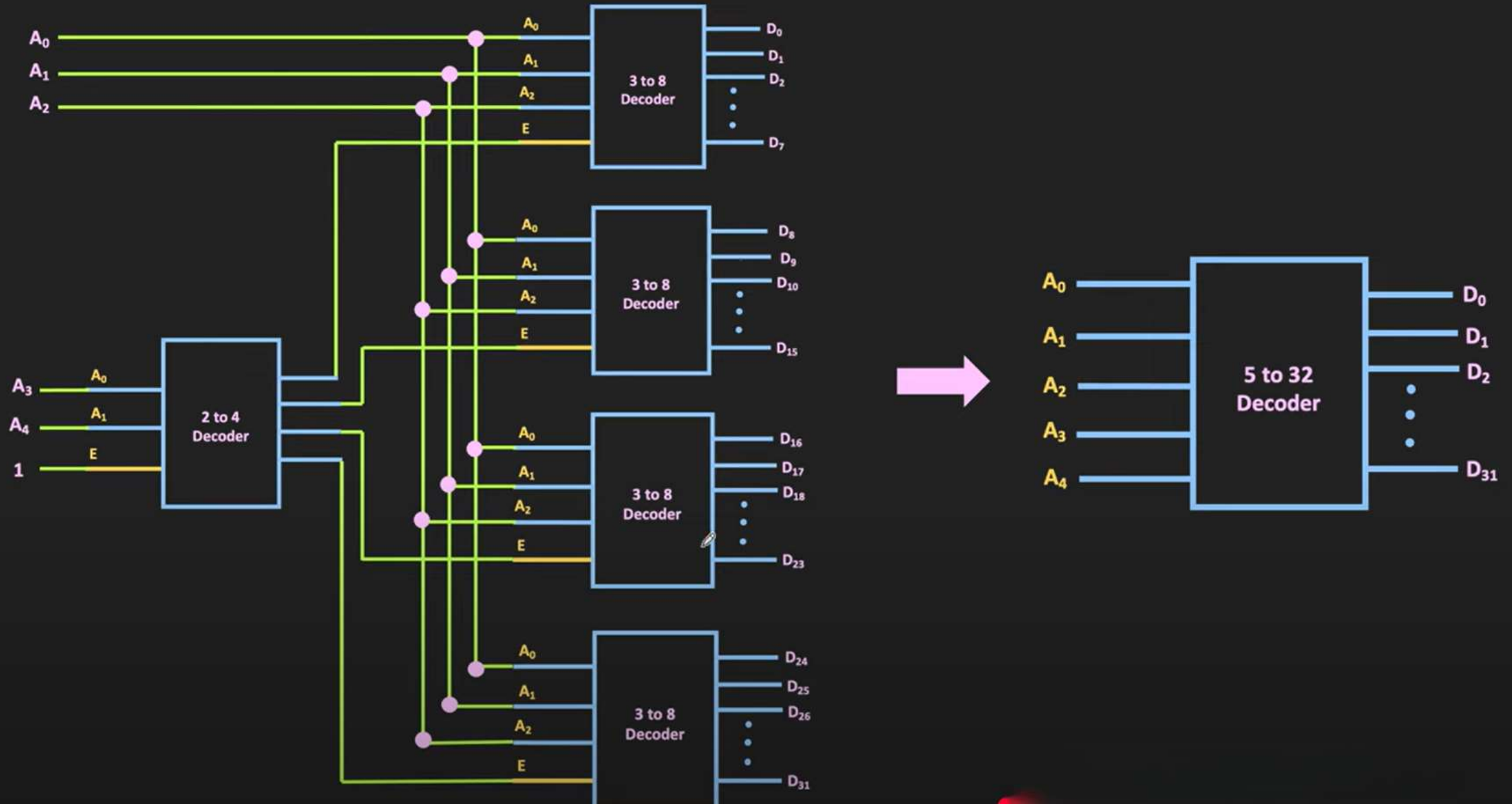
$$D = D_1 + D_3 + D_5 + D_7 + D_9$$



4 to 16 Decoder using 3 to 8 Decoders



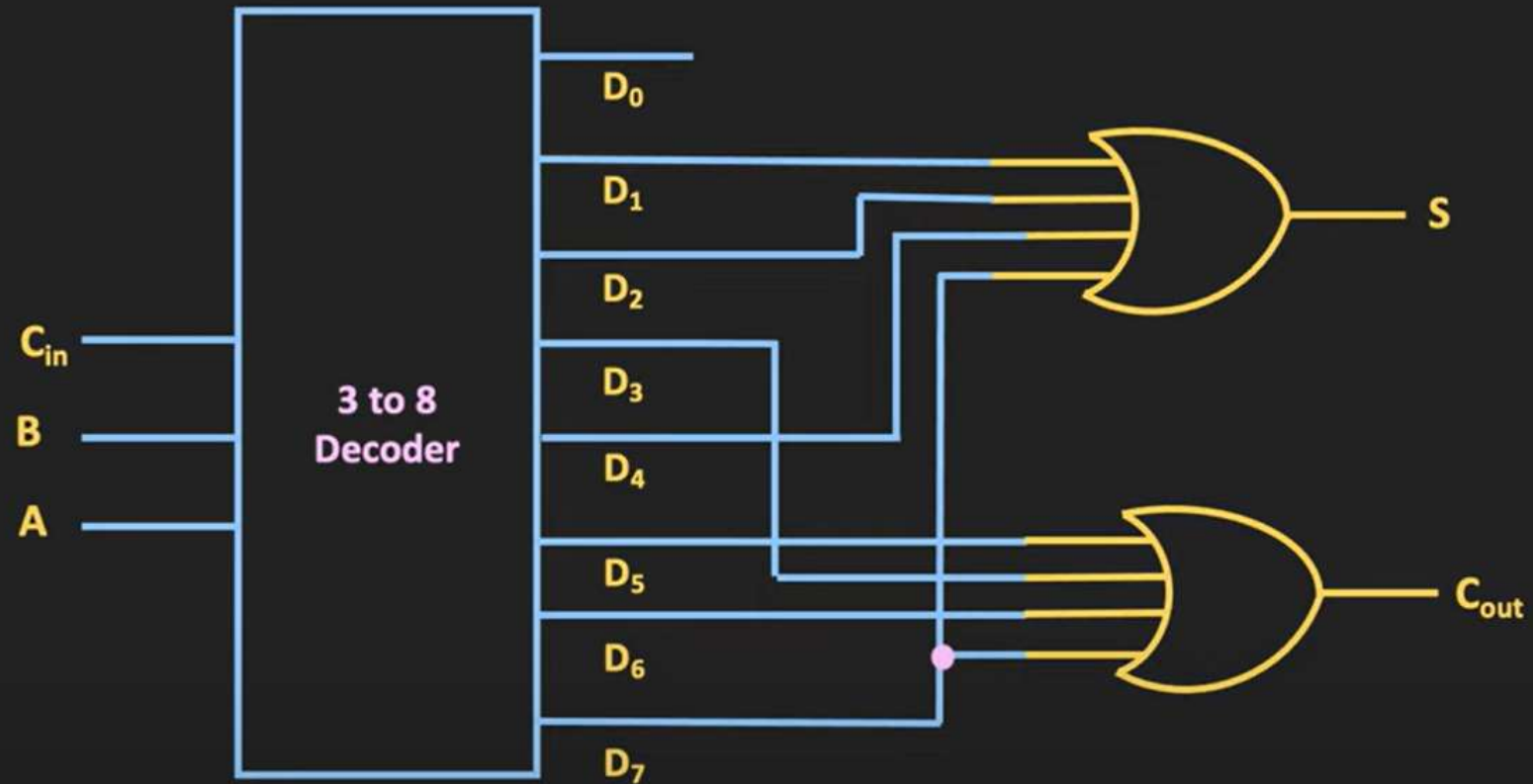
5 to 32 Decoder using 3 to 8 Decoders



Q) Implement full adder circuit using Decoder

Logic Circuit Implementation

A	B	C _{in}	Sum	C _{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

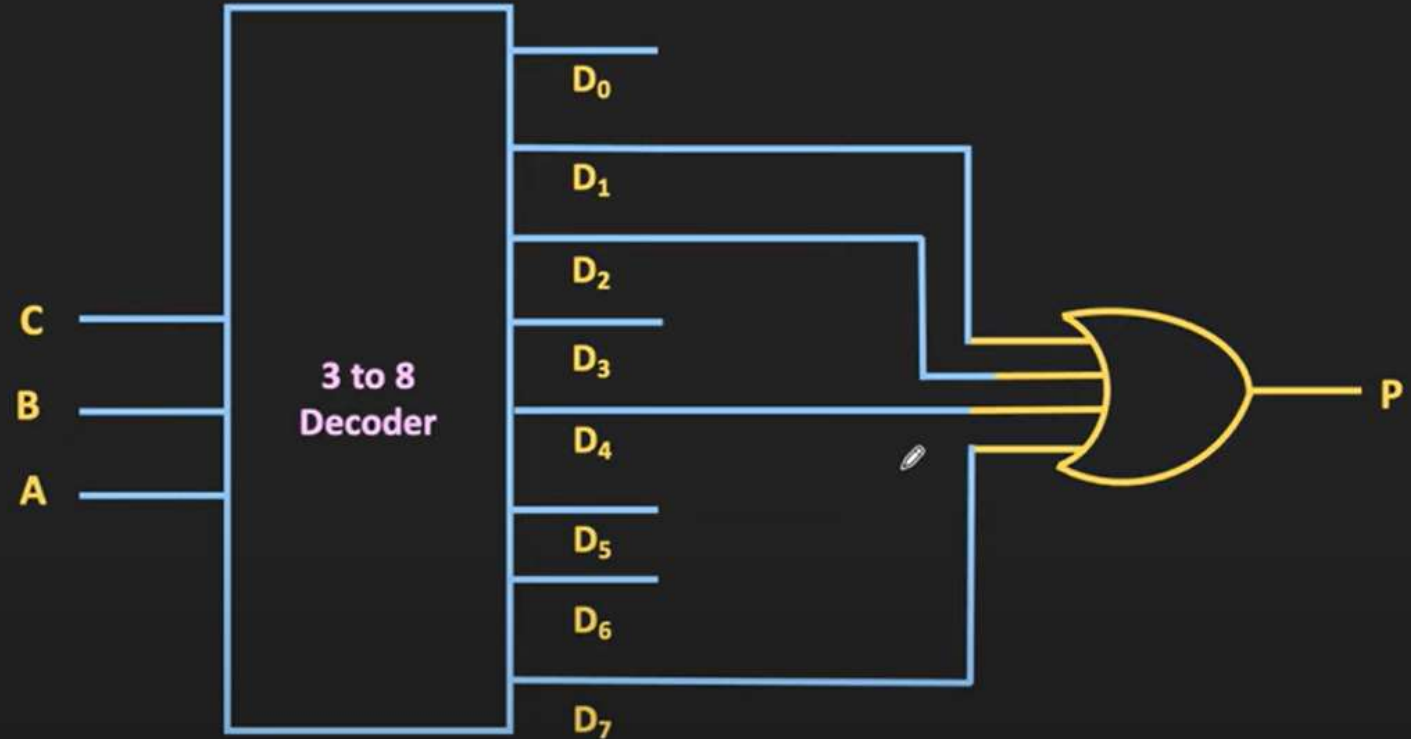


Q) Implement 3 bit even parity checker using decoder

Logic Circuit Implementation

Even Parity Checker

A	B	C	P
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1



Q) Implement $F = (0, 1, 2, 5, 6, 7)$ using decoder

Logic Circuit Implementation

A	B	C	Y
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

